CSA0614- TOPIC 3 – DAA

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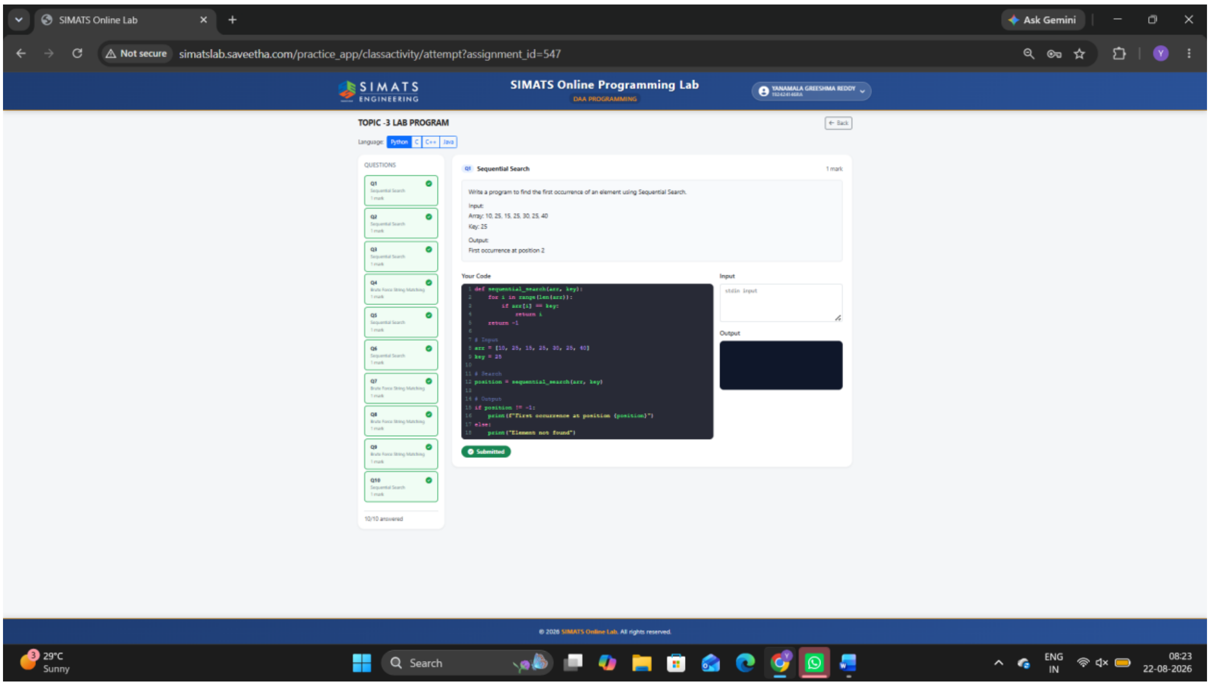

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Q2 Sequential Search

1 mark

Implement Sequential Search for an unsuccessful search.

Input:
Array: 5, 10, 15, 20, 25, 30, 35
Key: 18

Output:
Element not found
Number of comparisons = 7

Your Code

```python
def sequential_search(arr, key):
    comparisons = 0
    for i in range(len(arr)):
        comparisons += 1
        if arr[i] == key:
            return i, comparisons
    return -1, comparisons

# Input
arr = [5, 10, 15, 20, 25, 30, 35]
key = 18

index, comparisons = sequential_search(arr, key)

if index == -1:
    print("Element not found")
else:
    print(f"Element found at index {index}")

print(f"Number of comparisons = {comparisons}")
```

Input

stdin input

Output

```
Element not found
Number of comparisons = 7
```

Submitted

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Q3 Sequential Search

1 mark

Write a program to count:
- Total comparisons
- Total matches
- Total mismatches

during Sequential Search.

Input:
Array: 3, 6, 9, 12, 15, 18, 21
Key: 15

Your Code

```python
def sequential_search(arr, key):
    comparisons = 0
    matches = 0
    mismatches = 0

    for i in range(len(arr)):
        comparisons += 1
        if arr[i] == key:
            matches += 1
        else:
            mismatches += 1

    return comparisons, matches, mismatches

# Input array and key
arr = [3, 6, 9, 12, 15, 18, 21]
key = 15

comparisons, matches, mismatches = sequential_search(arr, key)

print("Total comparisons:", comparisons)
print("Total matches:", matches)
print("Total mismatches:", mismatches)
```

Input

stdin input

Output

```
Total comparisons: 7
Total matches: 1
Total mismatches: 6
```

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QUESTIONS

- Q1 Sequential Search 1 mark
- Q2 Sequential Search 1 mark
- Q3 Sequential Search 1 mark
- Q4 Brute Force String Matching 1 mark
- Q5 Sequential Search 1 mark
- Q6 Sequential Search 1 mark
- Q7 Brute Force String Matching 1 mark
- Q8 Brute Force String Matching 1 mark
- Q9 Brute Force String Matching 1 mark
- Q10 Sequential Search 1 mark

Q4 Brute Force String Matching

1 mark

Implement the Brute Force String Matching Algorithm to search for a pattern in a given text.

Input:
- Text = "AABAACAADAABAABA"
- Pattern = "AABA"

Output:
- Position(s) where the pattern occurs.
- Total number of comparisons.

Your Code

```python
def brute_force_search(text, pattern):
    n, m = len(text), len(pattern)
    comparisons = 0
    positions = []

    for i in range(n - m + 1):
        j = 0
        while j < m:
            comparisons += 1
            if text[i + j] != pattern[j]:
                break
            j += 1
        if j == m:
            positions.append(i)

    return positions, comparisons

text = "AABAACAADAABAABA"
pattern = "AABA"

pos, comp = brute_force_search(text, pattern)
print("Positions:", pos)
print("Total comparisons:", comp)
```

Input

stdin input

Output

```
Positions: [0, 9, 12]
Total comparisons: 30
```

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QUESTIONS

- Q1 Sequential Search 1 mark
- Q2 Sequential Search 1 mark
- Q3 Sequential Search 1 mark
- Q4 Brute Force String Matching 1 mark
- Q5 Sequential Search 1 mark
- Q6 Sequential Search 1 mark
- Q7 Brute Force String Matching 1 mark
- Q8 Brute Force String Matching 1 mark
- Q9 Brute Force String Matching 1 mark
- Q10 Sequential Search

Q5 Sequential Search

1 mark

Implement Sequential Search for string data.

Input:
Names:
Anu, Bala, Charan, Deepa, Esha, Farhan

Search Name:
Deepa

Output:
Name found at position 4

Your Code

```python
names = ["Anu", "Bala", "Charan", "Deepa", "Esha", "Farhan"]
search_name = "Deepa"

found = False
for i in range(len(names)):
    if names[i] == search_name:
        found = True
        print(f"Name found at position {i+1}")
        break

if not found:
    print("Name not found")
```

Input

stdin input

Output

```
Name found at position 4
```

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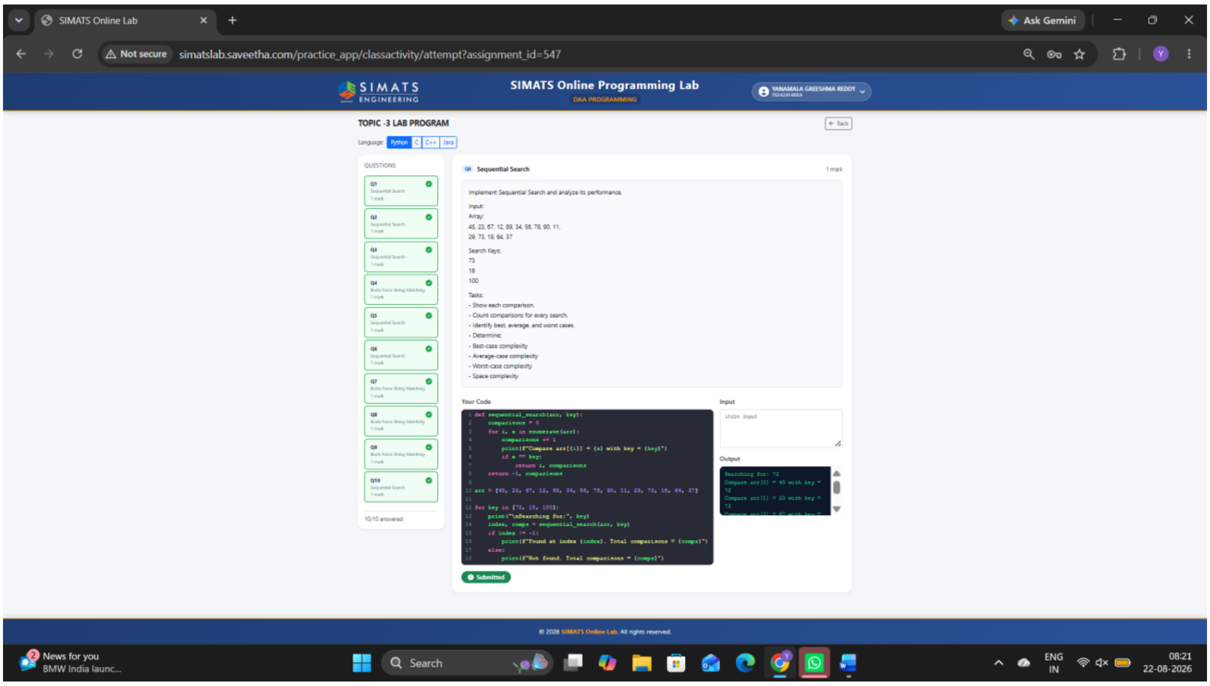
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Language: Python C C++ Java
QUESTIONS
Q6 Sequential Search
1 mark
Implement Sequential Search and analyze its performance.
Input:
Array:
45, 23, 67, 12, 89, 34, 56, 78, 90, 11,
29, 73, 18, 64, 37
Search Keys:
73
18
100
Tasks:
- Show each comparison.
- Count comparisons for every search.
- Identify best, average, and worst cases.
- Determine:
- Best-case complexity
- Average-case complexity
- Worst-case complexity
- Space complexity
Your Code
def sequential_search(arr, key):
comparisons = 0
for i, x in enumerate(arr):
comparisons += 1
print(f"Compare arr[{i}] = {x} with key = {key}")
if x == key:
return i, comparisons
return -1, comparisons
arr = [45, 23, 67, 12, 89, 34, 56, 78, 90, 11, 29, 73, 18, 64, 37]
for key in [73, 18, 100]:
print("\nSearching for:", key)
index, comps = sequential_search(arr, key)
if index != -1:
print(f"Found at index {index}. Total comparisons = {comps}")
else:
print(f"Not found. Total comparisons = {comps}")
Input
stdin input
Output
Searching for: 73
Compare arr[0] = 45 with key = 73
Compare arr[1] = 23 with key = 73
Submitted
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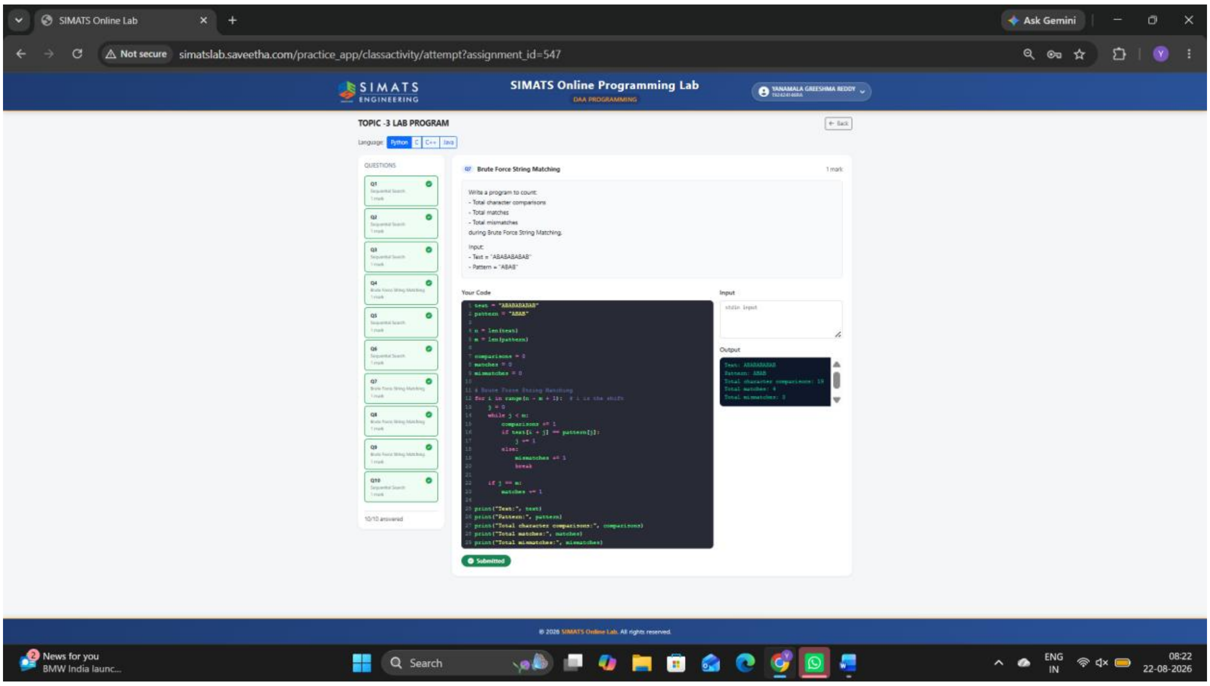
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TOPIC -3 LAB PROGRAM
Back
Language: Python C C++ Java
QUESTIONS
Q7 Brute Force String Matching
1 mark
Write a program to count:
- Total character comparisons
- Total matches
- Total mismatches
during Brute Force String Matching.
Input:
- Text = "ABABABABAB"
- Pattern = "ABAB"
Your Code
text = "ABABABABAB"
pattern = "ABAB"
n = len(text)
m = len(pattern)
comparisons = 0
matches = 0
mismatches = 0
# Brute Force String Matching
for i in range(n - m + 1): # i is the shift
j = 0
while j < m:
comparisons += 1
if text[i + j] == pattern[j]:
j += 1
else:
mismatches += 1
break
if j == m:
matches += 1
print("Text:", text)
print("Pattern:", pattern)
print("Total character comparisons:", comparisons)
print("Total matches:", matches)
print("Total mismatches:", mismatches)
Input
stdin input
Output
Text: ABABABABAB
Pattern: ABAB
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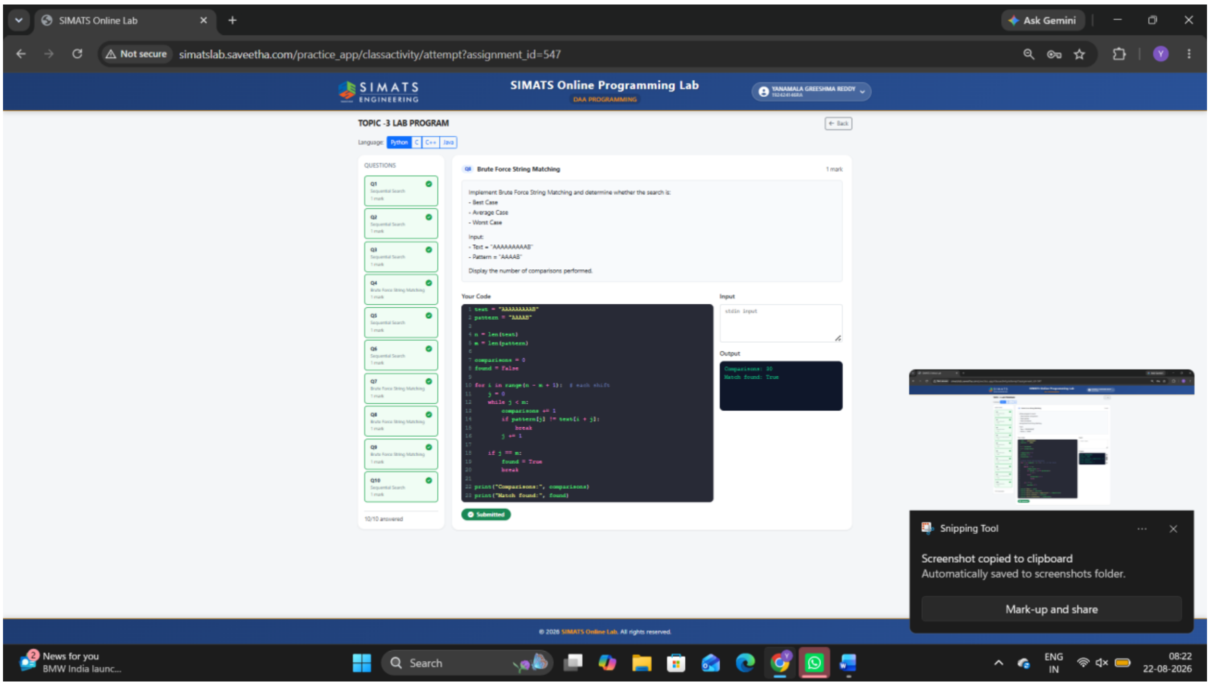

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Language: Python C C++ Java
QUESTIONS
Q8 Brute Force String Matching
1 mark
Implement Brute Force String Matching and determine whether the search is:
- Best Case
- Average Case
- Worst Case
Input:
- Text = "AAAAAAAAAB"
- Pattern = "AAAAB"
Display the number of comparisons performed.
Your Code
text = "AAAAAAAAAB"
pattern = "AAAAB"
n = len(text)
m = len(pattern)
comparisons = 0
found = False
for i in range(n - m + 1): # each shift
j = 0
while j < m:
comparisons += 1
if pattern[j] != text[i + j]:
break
j += 1
if j == m:
found = True
break
print("Comparisons:", comparisons)
print("Match found:", found)
Input
stdin input
Output
Comparisons: 30
Match found: True
Submitted
10/10 answered
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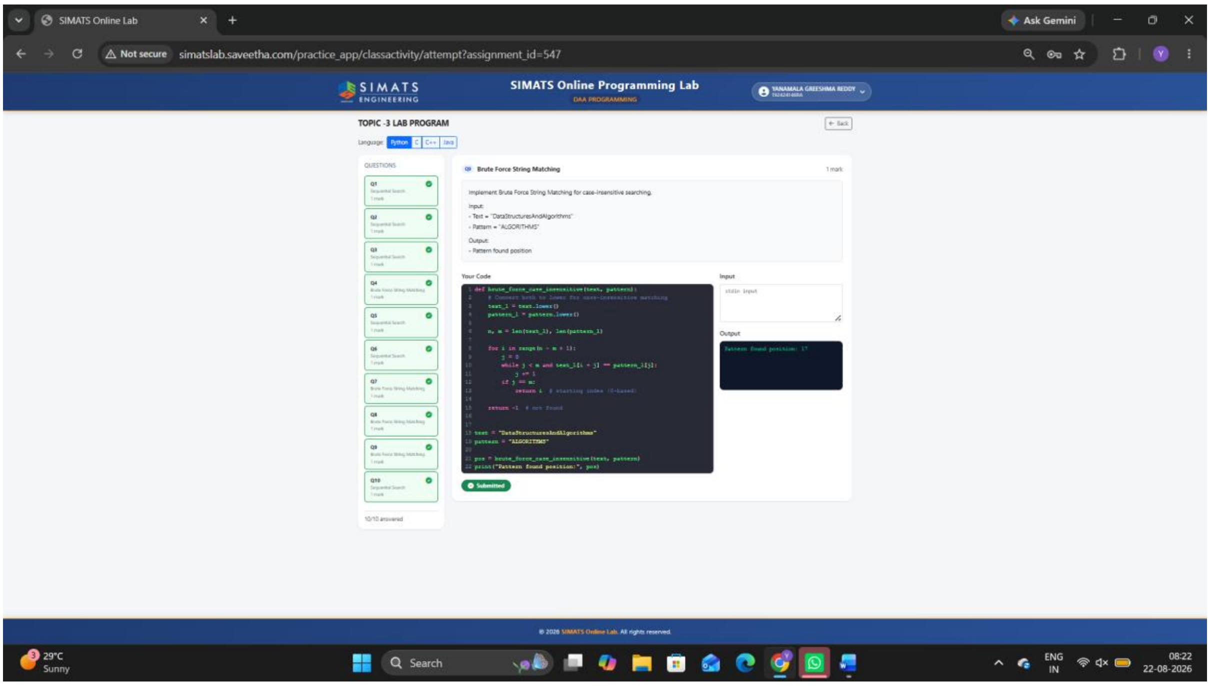

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Language: Python C C++ Java
QUESTIONS
Q9 Brute Force String Matching
1 mark
Implement Brute Force String Matching for case-insensitive searching.
Input:
- Text = "DataStructuresAndAlgorithms"
- Pattern = "ALGORITHMS"
Output:
- Pattern found position
Your Code
def brute_force_case_insensitive(text, pattern):
text_l = text.lower()
pattern_l = pattern.lower()
n, m = len(text_l), len(pattern_l)
for i in range(n - m + 1):
j = 0
while j < m and text_l[i + j] == pattern_l[j]:
j += 1
if j == m:
return i
return -1
text = "DataStructuresAndAlgorithms"
pattern = "ALGORITHMS"
pos = brute_force_case_insensitive(text, pattern)
print("Pattern found position:", pos)
Input
stdin input
Output
Pattern found position: 17
Submitted
10/10 answered
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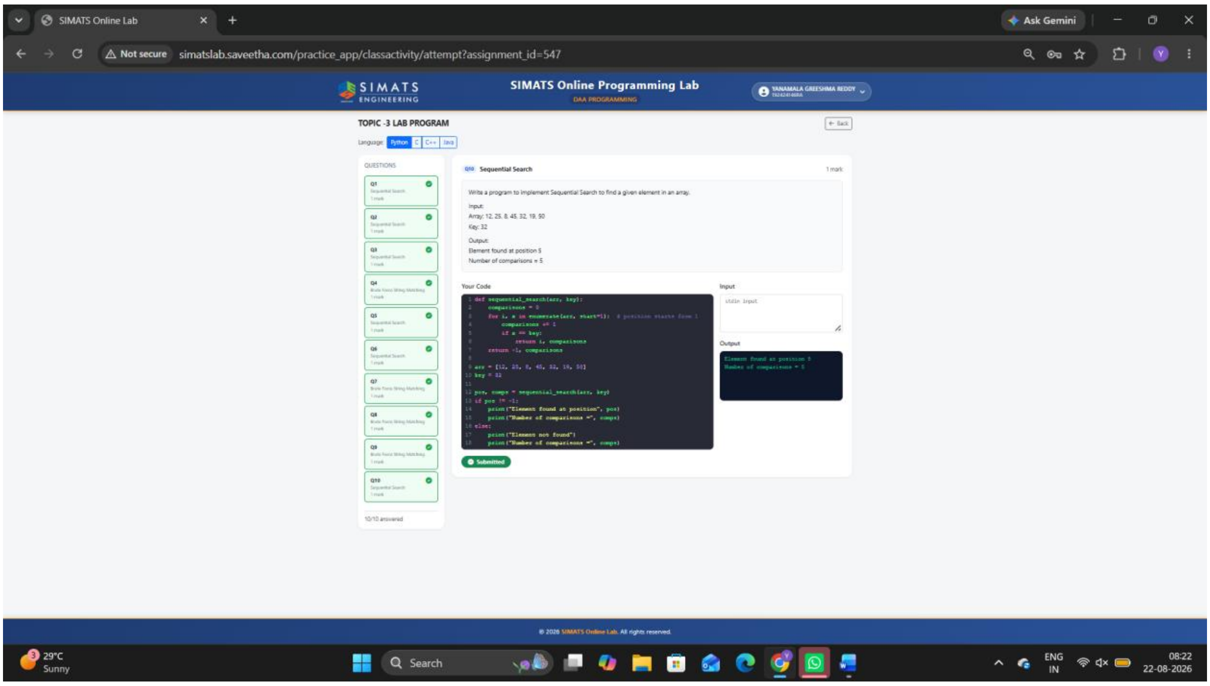
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TOPIC -3 LAB PROGRAM
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Language: Python C C++ Java
QUESTIONS
Q10 Sequential Search
1 mark
Write a program to implement Sequential Search to find a given element in an array.
Input:
Array: 12, 25, 8, 45, 32, 19, 50
Key: 32
Output:
Element found at position 5
Number of comparisons = 5
Your Code
def sequential_search(arr, key):
comparisons = 0
for i, x in enumerate(arr, start=1):
comparisons += 1
if x == key:
return i, comparisons
return -1, comparisons
pos, comps = sequential_search(arr, key)
if pos != -1:
print("Element found at position", pos)
print("Number of comparisons =", comps)
else:
print("Element not found")
print("Number of comparisons =", comps)
Input
stdin input
Output
Element found at position 5
Number of comparisons = 5
Submitted
10/10 answered
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